

"One City. Multiple AI Guardians. Infinite Possibilities."

Agentic Arena 2026 - Ideathon - Open Innovation - Case Study 01

Submitted by **Harshita Chokhani**

SUBMISSION LINKS	
SOURCE CODE REPOSITORY	https://github.com/HarshitaChokhani/GuardianGrid
DEMO VIDEO (Google Drive)	https://drive.google.com/drive/folders/1hF-nDw6hEbHkdnfdLhDOIEm99LrbkxuX?usp=drive_link
PROJECT DOCUMENTATION	This PDF (sections 1-12) + original proposal appended (Appendix A)
LIVE PROTOTYPE	Run locally - clone repo, then open index.html (no build step required)

Note: please set the Google Drive video to "Anyone with the link can view" so judges can open it.

1 Executive Summary

GuardianGrid is a multi-agent AI platform that reinvents how citizens interact with city services. Instead of juggling dozens of disconnected government portals and helplines, a citizen files **one** request - a photo, a line of text, or voice - and a team of autonomous AI agents collaborate to classify it, decide its priority, route it to the correct municipal department, dispatch a field crew, and keep the citizen updated until it is resolved. The submitted prototype is a fully clickable, end-to-end product spanning **three connected experiences**: a citizen grievance portal, an explainable AI Resolution Engine, and a government / officer console - tied together by a shared context store so every action on one side is reflected on the other.

2 Problem Statement

Modern cities still run on fragmented, manual systems. Citizens navigate dozens of siloed portals; complaint resolution is slow and opaque; departments operate in isolation; emergencies are slowed by congestion; and authorities lack predictive insight. With **68% of the world's population projected to live in cities by 2050 (UN)**, single-channel, manual service models simply cannot scale - coordination has to become autonomous.

3 The Solution - A Multi-Agent Architecture

Not one chatbot, but a collaborative ecosystem of specialised agents that share context and make coordinated decisions on their own. A Supervisor agent orchestrates a fleet of specialised guardians over a single shared memory, so responses stay coordinated and are never duplicated.

- **Traffic Guardian** - predicts congestion, suggests alternate routes, clears emergency corridors.
- **Infrastructure / Utility Guardian** - detects failures, files maintenance tickets, dispatches repair crews.
- **Emergency Guardian** - coordinates ambulance / police / fire and secures the fastest route.
- **Government Guardian** - registers complaints, allocates departments, issues reference IDs.
- **Analytics Guardian** - detects recurring issues, builds heatmaps and forecasts infrastructure problems.
- **Shared memory & orchestration** - every agent communicates over one context store in real time.

4 What Was Built (Key Features)

Citizen Grievance Portal	Phone/OTP login; report wizard (photo upload -> AI auto-classification -> State/City/Ward with AI locality verification -> review). India-format complaint IDs (e.g. PB-LDH-2026-004822), SLA, and a live tracking timeline.
AI Resolution Engine	Per-complaint, explainable pipeline of 6 agents that reason about the actual report in plain language, each making a decision with a confidence score and a stated rationale. Headline metric: "6 autonomous decisions - 0 human actions".
Government / Officer Console	Department inbox where complaints land auto-routed, with AI triage shown to the officer. Officers advance status (Acknowledge -> Assign -> In Progress -> Resolved); updates reflect instantly on the citizen's tracker - closing the loop.
Cinematic City Mission Control	A NASA-style command room with a living digital-twin city, orbiting AI core, live reasoning feed and command terminal - the macro view of autonomous city response.
Nationwide coverage	All 28 states + 8 union territories with ~600 cities/districts and real municipal-body routing (BMC, BBMP, MCD, GHMC, etc.).

5 How the AI Agents Work

When a citizen submits a report, the Supervisor activates six specialised agents in sequence. Each reasons over the real input and emits one explainable decision:

Agent	Job	Example decision (with rationale)
Vision	Understands the photo	"I can see a burst pipeline" -> category Water Leakage (94%)
Triage	Sets priority	"HIGH - a hospital is 240 m away, so it is a safety risk"
Pattern	Duplicate / trend	"Found 2 similar reports in this ward - linking as a recurring hotspot"
Routing	Department & body	"Water issues here are Jal Board under Ludhiana MC - routing there"
Coordination	Officer & crew	"Assigned to Er. Ravi Kumar; crew dispatched; ETA 8 hrs"
Comms	Citizen updates	"SMS + live tracking link sent - I will update you at each step"

6 Agent Autonomy & Decision-Making

- **No human in the routing loop** - category, priority, department and crew are decided autonomously.
- **Every decision is explainable** - shown with a confidence score and a one-line "why", on both the citizen and officer views.
- **Self-coordinating agents** - they read and write a shared memory store rather than waiting on a dispatcher.
- **Human-in-the-loop governance** - officers can still review, override or reassign, keeping humans in control of outcomes.

7 Technical Architecture & Stack

Citizen (voice / text / image) -> React front-end -> FastAPI gateway -> Supervisor agent -> LangGraph orchestration -> specialised agents (Traffic, Utility, Emergency, Government, Analytics) -> ChromaDB shared memory -> OpenAI reasoning -> Maps API -> Notification service.

Submitted prototype: a dependency-free front-end implementation in pure **HTML, CSS and JavaScript** (index.html, styles.css, app.js) - no build step, runs offline by opening index.html. All animations, the multi-agent pipeline visualisation, classification, routing and the citizen/officer state-sharing are functional in the browser. The architecture above describes the intended production stack.

8 Real-World Impact

Challenge	GuardianGrid solution	Expected outcome
Slow complaint tracking	Automated routing & live status	48 hours -> ~12 minutes
Emergency coordination	Multi-agent collaboration	18 minutes -> ~7 minutes
15+ government portals	One unified AI assistant	Single front door for citizens
Infrastructure failures	Predictive monitoring	Lower maintenance cost

9 Scalability & Future Scope

NEAR TERM	MID TERM	LONG TERM
-----------	----------	-----------

IoT sensor integration CCTV analytics Smart public transport	Digital twin cities Drone-based monitoring Disaster response	AI urban planning Sustainability monitoring Carbon-footprint tracking
--	--	---

1 Evaluation Criteria - How GuardianGrid Delivers

0

	Wt	How it is addressed
ity	25%	Six specialised agents process every real complaint end-to-end: vision, triage, de-duplication, routing, dispatch and citizen co
ivity	20%	AI-to-AI collaboration over shared memory; duplicate / hotspot detection; a dual citizen + government experience that closes t
itation	20%	Functional multi-agent pipeline, classification and routing across 36 states; clean, dependency-free front-end; documented pro
Decision-Making	15%	Autonomous priority and routing; every decision shown with confidence + rationale; "0 human actions" headline.
& Practicality	10%	India-specific municipal routing, complaint IDs, SLAs; officer console proves the complaint reaches a real department and is r
resentation	10%	One cinematic, narratable story across both sides; premium, animated, human-centred UI.

1 How to Run the Prototype

1

1. Clone the repository:

```
git clone https://github.com/HarshitaChokhani/GuardianGrid.git
```

2. Open **index.html** directly in a modern browser (Chrome recommended), or serve it:

```
cd GuardianGrid && python3 -m http.server 4173 # then open http://localhost:4173
```

No build step, no dependencies. Best viewed full-screen on desktop.

1 Demo Walkthrough (90 seconds)

2

- **Report an Issue** -> upload a photo (or pick a sample) -> AI auto-classifies the category & department -> confirm State/City/Ward (AI verifies the local body) -> submit -> receive a complaint ID.
- **Watch AI handle it** -> the 6-agent AI Resolution Engine reasons through the complaint live ("6 autonomous decisions - 0 human actions").
- **Government / Officer login** -> the complaint is sitting in the correct department inbox with AI triage -> click **Mark Resolved**.
- **Back to the citizen** -> My Complaints now shows **Resolved** - the loop is closed.

"GuardianGrid envisions a future where every city is empowered by collaborative AI agents that anticipate challenges, coordinate solutions, and create safer, smarter, and more connected communities."

Appendix A (following pages): original project proposal - GuardianGrid_Proposal.pdf

GuardianGrid

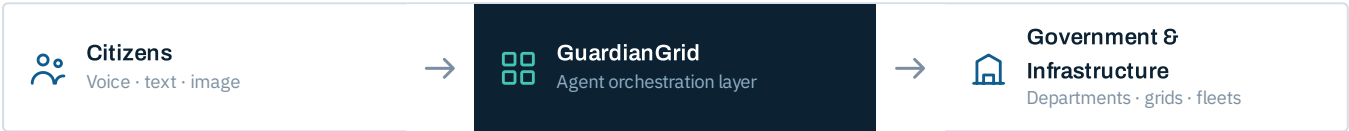
AI-Powered Multi-Agent Smart City Assistant

One City. Multiple AI Guardians. Infinite Possibilities.

Project Overview

01

GuardianGrid is a next-generation **multi-agent AI platform** that reinvents how citizens interact with city services. Rather than juggling separate portals and helplines, a citizen makes one request and a team of autonomous agents **collaborate to resolve civic issues, optimise operations, and improve engagement** in real time. In effect, it is an intelligent operating layer sitting between the people of a city and everything that runs it.



Problem Statement

02

Modern cities still run on fragmented systems that slow everything down:

- Citizens navigate dozens of disconnected government portals.
 - Departments operate in isolated, siloed systems.
 - Utility failures are reported and triaged manually.
 - Authorities lack predictive insight for proactive planning.
- Complaint resolution is slow and lacks transparency.
 - Congestion lengthens critical emergency response times.
 - Citizens receive delayed, inconsistent updates.
 - Rapid urban growth demands far smarter governance.

68%

of the world's population is projected to live in cities by 2050 (UN). Single-channel, manual service models simply cannot scale to that load — coordination has to become autonomous.

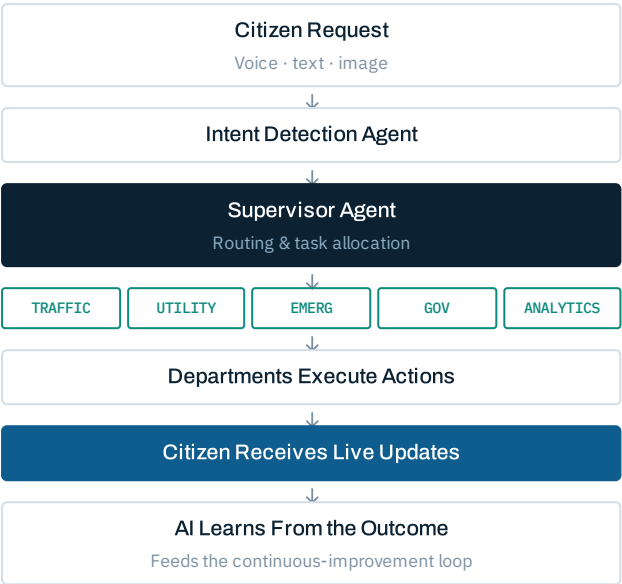
Proposed Solution

03

Not one chatbot, but a **collaborative ecosystem of specialised agents** that share context and make coordinated decisions on their own.

 Traffic Intelligence Predicts congestion, suggests alternate routes, and clears the way for emergency vehicles.	 Civic Services Handles garbage collection, road damage, streetlight failures and water-leakage complaints.	 Utility Monitoring Detects infrastructure failures, tracks power and water, and predicts maintenance needs.
 Emergency Response Coordinates ambulance, police and fire, finds the nearest responders, and streams live updates.	 Government Services Guides citizens through certificates, licences, taxes, schemes and appointment booking.	 City Analytics Builds dashboards, surfaces recurring issues, and forecasts future infrastructure problems.

Shared memory & orchestration — every agent communicates continuously over one shared context store, coordinating decisions in real time instead of working in isolation.



- 01 **One request, any format.** Voice, text or a photo of the issue enters through a single front door.
- 02 **Intent is understood** and structured before anything is routed anywhere.
- 03 **The Supervisor orchestrates**, allocating the task to the right specialised agent or agents.
- 04 **Agents act in parallel**, sharing context so the response stays coordinated, never duplicated.
- 05 **Departments execute** and the citizen is kept informed with live status throughout.
- 06 **Every outcome trains the system**, closing the loop and sharpening the next response.

Expected Impact

Challenge	GuardianGrid solution	Expected outcome
Traffic delays	AI rerouting & signal priority	Faster emergency response
Complaint tracking	Automated routing & live status	Reduced resolution time
Government services	Unified AI assistant	Better citizen experience
Infrastructure failures	Predictive monitoring	Lower maintenance cost
Emergency coordination	Multi-agent collaboration	Faster rescue operations

Innovation Highlights

- + Multi-agent AI architecture
- + Predictive analytics
- + Shared memory between agents
- + Autonomous decision making
- + Human-in-the-loop governance
- + Scalable cloud architecture
- + AI-to-AI collaboration
- + Real-time civic intelligence
- + Citizen-centric design

Future Scope

NEAR TERM	MID TERM	LONG TERM
IoT sensor integration CCTV analytics Smart public transportation	Digital twin cities Drone-based monitoring Disaster response coordination	AI-powered urban planning Sustainability monitoring Carbon footprint tracking

Conclusion

GuardianGrid is not merely a chatbot. It is an intelligent network of autonomous AI agents that collaborate to make cities more **responsive, efficient, transparent and citizen-centric** — turning fragmented services into one coordinated intelligence that anticipates rather than reacts.

"GuardianGrid envisions a future where every city is empowered by **collaborative AI agents** that anticipate challenges, coordinate solutions, and create safer, smarter, and more connected communities."